

Database (Introduction)

What is Data?

- Data is a collection of raw facts, figures or symbols that represent information.

- This data will be stored in a Database.

Types $\begin{cases} \rightarrow \text{Structured Data} \\ \rightarrow \text{Unstructured Data} \end{cases}$

What is Database?

- A database is an organized collection of data that can be easily accessed, managed and updated.

- It stores the data in structured format; to support efficient retrieval & manipulation.

What is DBMS?

- DBMS is a software that helps store, manage and retrieve data from databases efficiently.

- DBMS stands for Database Management System.

Types of DBMS :

Types

DBMS Examples:

Relational

MySQL, PostgreSQL, Oracle, SQL Server

Non-Relational

MongoDB, Cassandra, Redis ...

Relational Database :

→ Relational Database data can be stored in the tables.

→ each table consists of rows & columns

→ each column as name & datatype

→ Each row will be treated as record. which formed by single or multiple columns.

primary key & Foreign key :-

Primary key :-

- Primary key is a unique identifier

for each record (row) in a table.

- It cannot be null & must be unique.
- Every table should have one primary key.

Eg:- In an "Employee" Table,

- emp_id is the primary key

Foreign key:-

- Foreign key is a reference to the primary key in another table.
- It is used to connect two tables.
- It helps maintain referential integrity (valid links b/w tables).

Eg:- In an "Project" Table,

- 'emp_id' in the project table is a foreign key referencing emp_id in employee table.

Non-Relational Database:-

A non-Relational Database is a type of DB, that doesn't use tables and rows like relational DB. Instead, it stores data in various formats

such as:

- Document
- key-value pair
- Wide-column stores
- Graph structures.

- Designed to handle large volume of Unstructured & semi-structured data.

Offers flexibility, scalability and is ideal for Big Data, Real-Time Analytics, and AI/ML applications

Types of Non-Relational DBs.

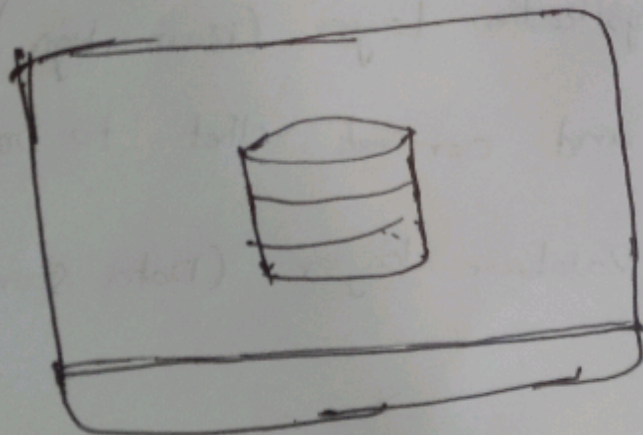
- Document DB
- Key-Value DB
- Wide-column DB
- Graph DB
- Search DB
- Vector DB.

Database Architectures in DBMS.

i) Single-Tier Architecture (1-Tier)

- The database and the user interface (application) are on the same system.
- Suitable for local, small, personal database.

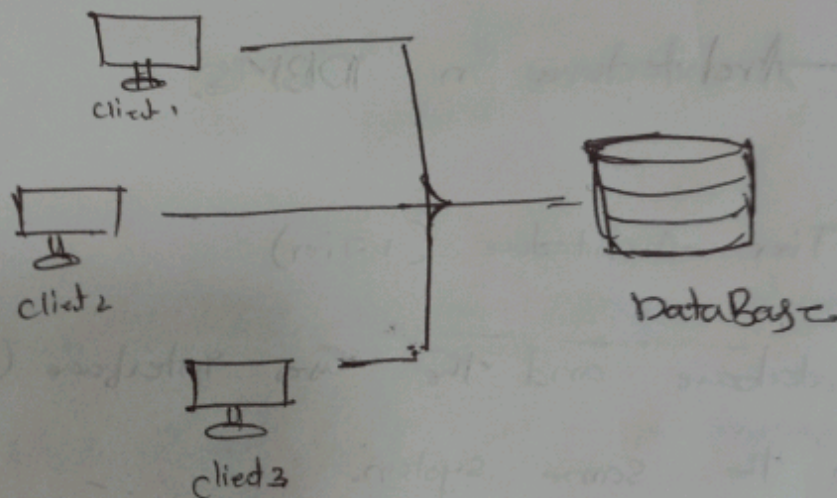
Eg:- MS Access on a single computer



2) Two-Tier Architecture (2-Tier)

- Client : sends request (UI or application)
- Server : stores the databases and processes queries (DBMS)

Eg:- MySQL + python Application



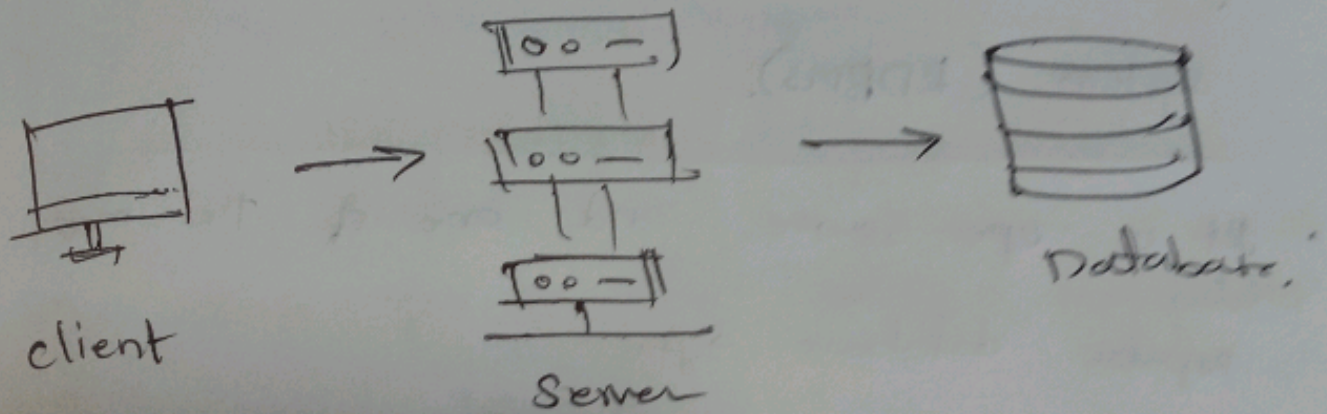
3) Three-Tier Architecture (3-Tier)

Tier - 1 : presentation layer (client/UI) - User Interface.

Tier - 2 : Application layer (Business logic) - process logic and connects client to DB.

Tier - 3 : Database layer (Data storage) - stores and retrieves data.

Ex-1- Web App (React frontend → Flask backend → PostgreSQL database)



MySQL

What is MySQL?

- MySQL is a relational database management system (RDBMS).
- It is open source and one of the most popular database sys.
- MySQL stores data in database, which contains tables.
- Each table organize data in rows & columns.
- It use SQL (Structured Query language) to perform operations like creating, reading, updating, and deleting data.
- It supports multiple users accessing the data securely at the same time.

What is SQL?

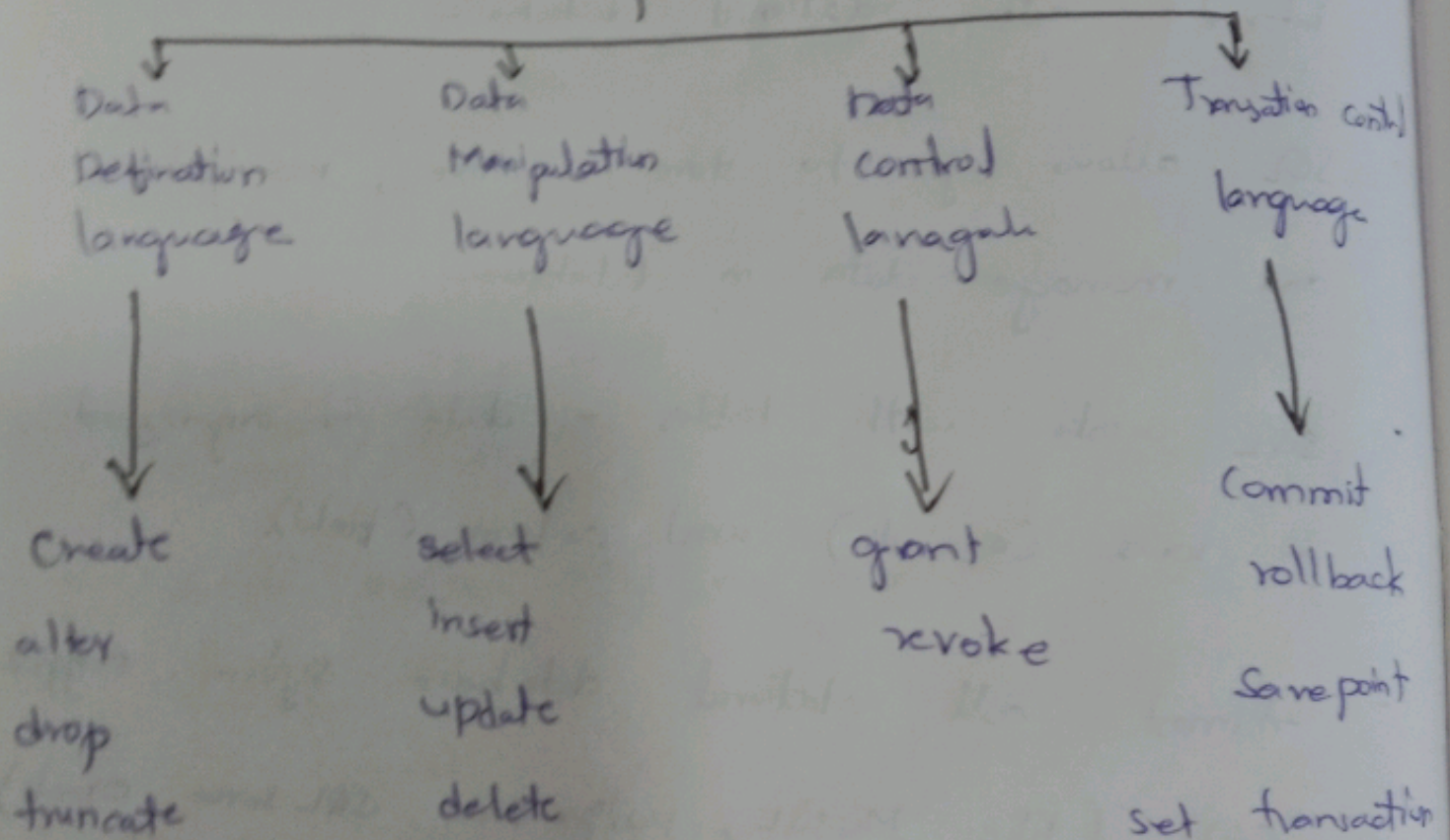
- SQL stands for Structured Query language.
- It is the standard language used to interact with relational database.
- SQL allows you to store, retrieve, manipulate, and manage data in databases.
- SQL works with tables - data is organized in rows (records) and columns (field).
- Almost all relational database systems support SQL (like MySQL, PostgreSQL, SQL server, Oracle)

Basic SQL Operations (CRUD)

Operations.	SQL Commands	Purpose
Create data	Insert Into	Add new records
Read data	SELECT	Retrieve data
Update data	UPDATE	Modify existing ^{records} data
Delete data	DELETE	Remove Records.

Types of SQL Commands.

SQL Language.



What is an Operator?

An operator in SQL is a symbol or keyword used in expression to perform:

- Arithmetic operations (e.g., +, -)
- Comparisons (e.g., =, !=)
- Logical checks (e.g., AND, OR)
- Condition negation (e.g., NOT)

→ These are mainly used in the "where" clause to filter records.

Types of SQL Operators:

1. Arithmetic Operators.

Used for basic mathematical operations.

- Addition +
- Subtraction -
- Multiplication *
- Division /
- Modulus %

Example:-

Select 10 + 20;

// output : 30

Select 12 % 5;

// output : 2.

2. Comparison Operators

Used to compare values:

- Equal to =
- Not equal to $\neq$
- Not equal to $\langle \rangle$ (alternative syntax)
- Greater than $>$
- Less than $<$
- Greater than ^{or equal to} $\geq$
- Less than or equal to $\leq$
- Not less than $\nless$
- Not Greater than $\ngtr$

Examples:

Select * from customers where Salary $>$ 5000;
Select * from customers where salary $\neq$ 2000;

3. Logical Operators

Used to combine multiple conditions

- AND - True if all conditions are true
- OR - True if any condition is true
- NOT - Reverse the results
- BETWEEN - True if value is within a specified range
- IN - True if value matches any in a given list.
- LIKE - True if value matches a pattern.
(wildcards = %, -)
- IS NULL - checks if value is null.
- EXISTS - True if a subquery returns any result.
- ALL - Compares a value to all values in a set.
- ANY - Compares a value to any value in a set.
- UNIQUE - checks for uniqueness of rows.

Example :-

Select * from customers where Age ≥ 25 and salary ≥ 8500 .

SQL - Expressions.

1. Boolean Expression.

Used in the "where" clause to match specific values.

Eg:-

```
select * from customers where salary = 6000;
```

2. Numeric Expressions.

Used to perform mathematical operations in queries.

Eg:-

```
select (15 + 6) as Addition;
```

3. Aggregate Functions (part of Numeric).

Functions like `sum()`, `avg()`, `count()` help in data calculation.

Eg:-

```
select count(*) as "Records" from customers;
```


Data Expressions:

Returns current system date & time.

eg:- Select current_Timestamp;